K-mer GO-split Checkpoint Error Analysis

This notebook performs error analysis for the three namespace-specific k-mer MLP checkpoints:

- `kmer_BPO_final.pt`
- `kmer_CCO_final.pt`
- `kmer_MFO_final.pt`

It follows the GO-split setup from `kmer_GO_split.ipynb`.

This notebook does **not** retrain the models.
It only reloads saved checkpoints, runs validation inference, and analyzes errors.

Important note:

You do **not** need `kmer_dropout_0.3.pt` for this analysis.
That file is an intermediate tuning checkpoint.
For GO-split error analysis, we use the final namespace-specific checkpoints only.

Step 0. Imports and Paths

In [1]:
```
import os
import json
import pickle
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.sparse import load_npz

import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from sklearn.metrics import f1_score

DATA_DIR = "data_processed_kmer"

CHECKPOINTS = {
    "BPO": "pt/kmer_BPO_final.pt",
    "CCO": "pt/kmer_CCO_final.pt",
    "MFO": "pt/kmer_MFO_final.pt",
}

OUTPUT_DIR = "kmer_error_analysis_outputs"
os.makedirs(OUTPUT_DIR, exist_ok=True)

print("DATA_DIR:", DATA_DIR)
```

```
print("CHECKPOINTS:", CHECKPOINTS)
print("OUTPUT_DIR:", OUTPUT_DIR)
```

```
DATA_DIR: data_processed_kmer
CHECKPOINTS: {'BPO': 'pt/kmer_BPO_final.pt', 'CCO': 'pt/kmer_CCO_final.pt',
'MFO': 'pt/kmer_MFO_final.pt'}
OUTPUT_DIR: kmer_error_analysis_outputs
```

Step 1. Load Processed K-mer Data

Why this step?

The `.pt` files only store model weights.

To evaluate prediction errors, we still need the validation features and labels.

In [2]:
```
X_train = np.load(os.path.join(DATA_DIR, "X_train.npy"))
X_val = np.load(os.path.join(DATA_DIR, "X_val.npy"))

Y_train = load_npz(os.path.join(DATA_DIR, "Y_train.npz"))
Y_val = load_npz(os.path.join(DATA_DIR, "Y_val.npz"))

with open(os.path.join(DATA_DIR, "meta.pkl"), "rb") as f:
    meta = pickle.load(f)

go_terms = meta["go_terms"].tolist() if hasattr(meta["go_terms"], "tolist")

print("X_train:", X_train.shape)
print("X_val:", X_val.shape)
print("Y_train:", Y_train.shape)
print("Y_val:", Y_val.shape)
print("Number of GO terms:", len(go_terms))

assert Y_train.shape[1] == len(go_terms)
assert Y_val.shape[1] == len(go_terms)
```

```
X_train: (112455, 400)
X_val: (28114, 400)
Y_train: (112455, 31454)
Y_val: (28114, 31454)
Number of GO terms: 31454
```

Step 2. Load GO Namespace Mapping

Why this step?

The final GO-split models were trained separately for BPO, CCO, and MFO.

We need to reconstruct the same column splits used during training.

In [3]:
```
TERMS_PATH = os.path.join("..", "cafa-5-protein-function-prediction", "Trair

if not os.path.exists(TERMS_PATH):
    raise FileNotFoundError(
```

```
        f"Cannot find train_terms.tsv at {TERMS_PATH}. "
        "Please update TERMS_PATH."
    )

terms_df = pd.read_csv(TERMS_PATH, sep="\t")

go_to_aspect = (
    terms_df[["term", "aspect"]]
    .drop_duplicates()
    .set_index("term")["aspect"]
    .to_dict()
)

bpo_idx = [i for i, go in enumerate(go_terms) if go_to_aspect.get(go) == "BP
cco_idx = [i for i, go in enumerate(go_terms) if go_to_aspect.get(go) == "CC
mfo_idx = [i for i, go in enumerate(go_terms) if go_to_aspect.get(go) == "MF

namespace_indices = {
    "BPO": bpo_idx,
    "CCO": cco_idx,
    "MFO": mfo_idx,
}

namespace_go_terms = {
    "BPO": [go_terms[i] for i in bpo_idx],
    "CCO": [go_terms[i] for i in cco_idx],
    "MFO": [go_terms[i] for i in mfo_idx],
}

print("Total GO terms:", len(go_terms))
print("BPO labels:", len(bpo_idx))
print("CCO labels:", len(cco_idx))
print("MFO labels:", len(mfo_idx))
print("Check total:", len(bpo_idx) + len(cco_idx) + len(mfo_idx))
```

```
Total GO terms: 31454
BPO labels: 21279
CCO labels: 2957
MFO labels: 7218
Check total: 31454
```

Step 3. Split Validation Labels by Namespace

In [4]:

```
Y_val_namespace = {
    "BPO": Y_val[:, bpo_idx],
    "CCO": Y_val[:, cco_idx],
    "MFO": Y_val[:, mfo_idx],
}

Y_train_namespace = {
    "BPO": Y_train[:, bpo_idx],
    "CCO": Y_train[:, cco_idx],
    "MFO": Y_train[:, mfo_idx],
}
```

```
for ns in ["BPO", "CCO", "MFO"]:
    print(ns, "Y_train:", Y_train_namespace[ns].shape, "Y_val:", Y_val_names
```

```
BPO Y_train: (112455, 21279) Y_val: (28114, 21279)
CCO Y_train: (112455, 2957) Y_val: (28114, 2957)
MFO Y_train: (112455, 7218) Y_val: (28114, 7218)
```

Step 4. Define Dataset and MLP Architecture

Why this step?

Each `.pt` checkpoint stores weights only, so we rebuild the same MLP architecture before loading each checkpoint.

In [5]:
```
class ProteinKmerDataset(Dataset):
    def __init__(self, X, Y_sparse):
        self.X = X.astype(np.float32)
        self.Y = Y_sparse.tocsr()
        self.n_labels = self.Y.shape[1]

    def __len__(self):
        return self.X.shape[0]

    def __getitem__(self, idx):
        x = torch.from_numpy(self.X[idx])

        start = self.Y.indptr[idx]
        end = self.Y.indptr[idx + 1]
        pos_idx = self.Y.indices[start:end]

        y = torch.zeros(self.n_labels, dtype=torch.float32)
        y[pos_idx] = 1.0

        return x, y


class MLP(nn.Module):
    def __init__(self, input_dim, output_dim, hidden1=512, hidden2=256, drop
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden1),
            nn.BatchNorm1d(hidden1),
            nn.ReLU(),
            nn.Dropout(dropout),
            nn.Linear(hidden1, hidden2),
            nn.BatchNorm1d(hidden2),
            nn.ReLU(),
            nn.Dropout(dropout),
            nn.Linear(hidden2, output_dim)
        )

    def forward(self, x):
        return self.net(x)
```

```
def build_val_loader(X_val, Y_val_ns, batch_size=64):
    dataset = ProteinKmerDataset(X_val, Y_val_ns)
    loader = DataLoader(
        dataset,
        batch_size=batch_size,
        shuffle=False,
        num_workers=4,
        pin_memory=torch.cuda.is_available()
    )
    return dataset, loader


device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
input_dim = X_val.shape[1]

print("Device:", device)
print("input_dim:", input_dim)
```

```
Device: cpu
input_dim: 400
```

```
/global/home/users/yushanfu/esm_env/lib/python3.11/site-packages/torch/cuda/
__init__.py:180: UserWarning: CUDA initialization: The NVIDIA driver on your
system is too old (found version 12030). Please update your GPU driver by do
wnloading and installing a new version from the URL: http://www.nvidia.com/D
ownload/index.aspx Alternatively, go to: https://pytorch.org to install a Py
Torch version that has been compiled with your version of the CUDA driver.
(Triggered internally at /pytorch/c10/cuda/CUDAFunctions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
```

Step 5. Load Checkpoint and Infer Architecture

Why this step?

If the saved model uses hidden sizes different from the default, we infer the hidden dimensions from the checkpoint weight shapes.

In [6]:
```
def load_checkpoint_state_dict(path):
    if not os.path.exists(path):
        raise FileNotFoundError(
            f"Checkpoint not found: {path}. "
            "Please make sure the .pt file is in the current directory."
        )

    checkpoint = torch.load(path, map_location="cpu")

    if isinstance(checkpoint, dict) and "model_state_dict" in checkpoint:
        return checkpoint["model_state_dict"]
    elif isinstance(checkpoint, dict):
        return checkpoint
    else:
        raise ValueError("Unsupported checkpoint format.")
```

```
def infer_mlp_dims_from_state_dict(state_dict):
    if "net.0.weight" in state_dict and "net.4.weight" in state_dict:
        hidden1 = state_dict["net.0.weight"].shape[0]
        hidden2 = state_dict["net.4.weight"].shape[0]
    else:
        linear_weight_keys = [
            k for k in state_dict.keys()
            if k.endswith("weight") and len(state_dict[k].shape) == 2
        ]
        hidden1 = state_dict[linear_weight_keys[0]].shape[0]
        hidden2 = state_dict[linear_weight_keys[1]].shape[0]

    return int(hidden1), int(hidden2)


def build_model_from_checkpoint(checkpoint_path, input_dim, output_dim, devi
    state_dict = load_checkpoint_state_dict(checkpoint_path)
    hidden1, hidden2 = infer_mlp_dims_from_state_dict(state_dict)

    model = MLP(
        input_dim=input_dim,
        output_dim=output_dim,
        hidden1=hidden1,
        hidden2=hidden2,
        dropout=dropout
    ).to(device)

    model.load_state_dict(state_dict)
    model.eval()

    return model, {"hidden1": hidden1, "hidden2": hidden2, "dropout": dropou
```

Step 6. Run Validation Inference for Each Namespace

Why this step?

We only need a forward pass through the validation set.

No training is performed here.

In [7]:
```
@torch.no_grad()
def get_predictions(model, loader, device):
    model.eval()
    all_probs = []
    all_targets = []

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        logits = model(X_batch)
        probs = torch.sigmoid(logits).cpu().numpy()

        all_probs.append(probs)
        all_targets.append(Y_batch.numpy())
```

```
    y_prob = np.vstack(all_probs)
    y_true = np.vstack(all_targets)

    return y_prob, y_true


predictions = {}
model_configs = {}

for ns in ["BPO", "CCO", "MFO"]:
    print("=" * 80)
    print(f"Loading and evaluating {ns}")
    print("=" * 80)

    _, val_loader = build_val_loader(X_val, Y_val_namespace[ns], batch_size=

    model, inferred_config = build_model_from_checkpoint(
        checkpoint_path=CHECKPOINTS[ns],
        input_dim=input_dim,
        output_dim=Y_val_namespace[ns].shape[1],
        device=device,
        dropout=0.3
    )

    y_prob, y_true = get_predictions(model, val_loader, device)

    predictions[ns] = {
        "y_prob": y_prob,
        "y_true": y_true,
    }

    model_configs[ns] = inferred_config

    np.save(os.path.join(OUTPUT_DIR, f"{ns}_y_prob.npy"), y_prob)
    np.save(os.path.join(OUTPUT_DIR, f"{ns}_y_true.npy"), y_true)

    print(ns, "y_prob:", y_prob.shape, "y_true:", y_true.shape)
    print(ns, "inferred config:", inferred_config)
```

```
================================================================================
====
Loading and evaluating BPO
================================================================================
====
BPO y_prob: (28114, 21279) y_true: (28114, 21279)
BPO inferred config: {'hidden1': 512, 'hidden2': 256, 'dropout': 0.3}
================================================================================
====
Loading and evaluating CCO
================================================================================
====
CCO y_prob: (28114, 2957) y_true: (28114, 2957)
CCO inferred config: {'hidden1': 512, 'hidden2': 256, 'dropout': 0.3}
================================================================================
====
Loading and evaluating MFO
================================================================================
====
MFO y_prob: (28114, 7218) y_true: (28114, 7218)
MFO inferred config: {'hidden1': 512, 'hidden2': 256, 'dropout': 0.3}
```

Step 7. Threshold Sweep

Why this step?

Error analysis requires binary predictions.

We select the threshold that gives the best validation micro-F1 for each namespace.

In [8]:

```
BEST_THRESHOLDS = {
    "BPO": 0.75,
    "CCO": 0.75,
    "MFO": 0.85,
}

threshold_results = {}

for ns in ["BPO", "CCO", "MFO"]:
    y_prob = predictions[ns]["y_prob"]
    y_true = predictions[ns]["y_true"]

    selected_threshold = BEST_THRESHOLDS[ns]
    y_pred = (y_prob >= selected_threshold).astype(np.int8)

    micro = f1_score(y_true, y_pred, average="micro", zero_division=0)
    macro = f1_score(y_true, y_pred, average="macro", zero_division=0)

    threshold_results[ns] = {
        "selected_threshold": selected_threshold,
        "micro_f1": float(micro),
        "macro_f1": float(macro),
    }

    print("=" * 80)
```

```
    print(ns)
    print("Selected threshold:", selected_threshold)
    print("micro-F1:", micro)
    print("macro-F1:", macro)
```

```
==============================================================================
====
BPO
Selected threshold: 0.75
micro-F1: 0.10085526079523256
macro-F1: 0.00972265857580832
==============================================================================
====
CCO
Selected threshold: 0.75
micro-F1: 0.1463468320868456
macro-F1: 0.032351325050349294
==============================================================================
====
MFO
Selected threshold: 0.85
micro-F1: 0.15183777531219755
macro-F1: 0.017538097836723984
```

Step 8. Compute FP / FN / TP Error Summary

Why this step?

This identifies whether each namespace model is more likely to over-predict or miss GO terms.

In [9]:
```
def compute_error_analysis_for_namespace(ns, y_true, y_prob, threshold, go_t
    y_pred = (y_prob >= threshold).astype(np.int8)

    fp = (y_pred == 1) & (y_true == 0)
    fn = (y_pred == 0) & (y_true == 1)
    tp = (y_pred == 1) & (y_true == 1)

    fp_per_label = fp.sum(axis=0)
    fn_per_label = fn.sum(axis=0)
    tp_per_label = tp.sum(axis=0)

    true_count_per_label = y_true.sum(axis=0)
    pred_count_per_label = y_pred.sum(axis=0)

    summary = {
        "namespace": ns,
        "threshold": float(threshold),
        "total_tp": int(tp.sum()),
        "total_fp": int(fp.sum()),
        "total_fn": int(fn.sum()),
        "total_true_labels": int(y_true.sum()),
        "total_predicted_labels": int(y_pred.sum()),
        "avg_true_labels_per_protein": float(y_true.sum(axis=1).mean()),
```

```
        "avg_predicted_labels_per_protein": float(y_pred.sum(axis=1).mean())
    }

    label_error_df = pd.DataFrame({
        "namespace": ns,
        "go_term": go_terms_ns,
        "true_count": true_count_per_label,
        "pred_count": pred_count_per_label,
        "tp": tp_per_label,
        "fp": fp_per_label,
        "fn": fn_per_label,
    })

    label_error_df["fn_rate"] = label_error_df["fn"] / (label_error_df["true
    label_error_df["fp_rate"] = label_error_df["fp"] / (label_error_df["prec

    return summary, label_error_df


error_summaries = []
label_error_tables = {}

for ns in ["BPO", "CCO", "MFO"]:
    y_true = predictions[ns]["y_true"]
    y_prob = predictions[ns]["y_prob"]
    threshold = threshold_results[ns]["selected_threshold"]

    summary, label_error_df = compute_error_analysis_for_namespace(
        ns=ns,
        y_true=y_true,
        y_prob=y_prob,
        threshold=threshold,
        go_terms_ns=namespace_go_terms[ns],
    )

    error_summaries.append(summary)
    label_error_tables[ns] = label_error_df

    label_error_df.to_csv(
        os.path.join(OUTPUT_DIR, f"{ns}_per_label_error_table.csv"),
        index=False
    )

error_summary_df = pd.DataFrame(error_summaries)
display(error_summary_df)

error_summary_df.to_csv(
    os.path.join(OUTPUT_DIR, "namespace_error_summary.csv"),
    index=False
)
```

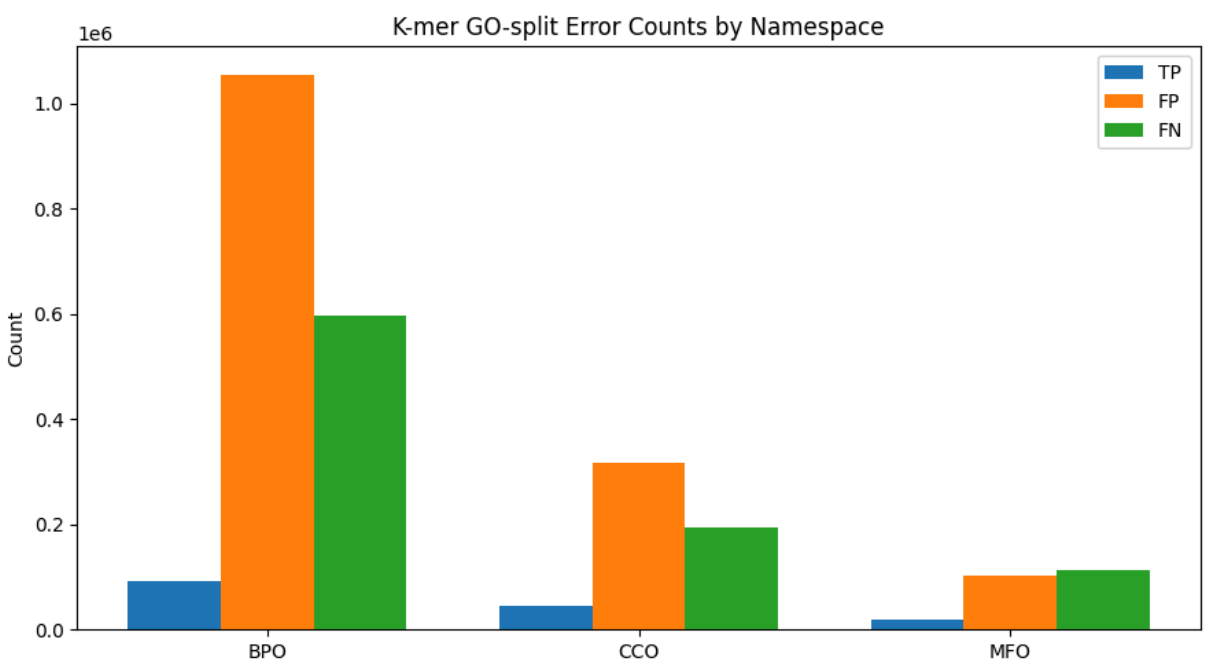
	namespace	threshold	total_tp	total_fp	total_fn	total_true_labels	total_predicted
0	BPO	0.75	92676	1055757	596693	689369	1
1	CCO	0.75	43881	317411	194512	238393	
2	MFO	0.85	19296	101504	114070	133366	

Step 9. Plot Error Counts by Namespace

In [10]:
```
x = np.arange(len(error_summary_df))
width = 0.25

plt.figure(figsize=(9, 5))
plt.bar(x - width, error_summary_df["total_tp"], width=width, label="TP")
plt.bar(x, error_summary_df["total_fp"], width=width, label="FP")
plt.bar(x + width, error_summary_df["total_fn"], width=width, label="FN")

plt.xticks(x, error_summary_df["namespace"])
plt.ylabel("Count")
plt.title("K-mer GO-split Error Counts by Namespace")
plt.legend()
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "namespace_error_counts.png"), dpi=200,
plt.show()
```



Step 10. Top False Negative GO Terms

Interpretation

False negatives are GO terms that are truly present but missed by the model.

In [11]:
```
for ns in ["BPO", "CCO", "MFO"]:
    print("=" * 80)
    print(f"Top False Negative GO terms: {ns}")
    print("=" * 80)

    top_fn = (
        label_error_tables[ns][label_error_tables[ns]["true_count"] > 0]
        .sort_values("fn", ascending=False)
        .head(15)
    )

    display(top_fn[["go_term", "true_count", "pred_count", "tp", "fn", "fn_r

    top_fn.to_csv(
        os.path.join(OUTPUT_DIR, f"{ns}_top_false_negative_terms.csv"),
        index=False
    )
```

```
================================================================================
Top False Negative GO terms: BPO
================================================================================
```

	go_term	true_count	pred_count	tp	fn	fn_rate
2653	GO:0008150	18271.0	1218	1037	17234	0.943243
3304	GO:0009987	12184.0	0	0	12184	1.000000
13362	GO:0065007	8153.0	61	46	8107	0.994358
11212	GO:0050789	7699.0	95	73	7626	0.990518
11216	GO:0050794	6617.0	98	62	6555	0.990630
11286	GO:0050896	6144.0	58	34	6110	0.994466
2654	GO:0008152	6106.0	311	182	5924	0.970193
14224	GO:0071704	5657.0	263	144	5513	0.974545
9329	GO:0044237	5084.0	169	83	5001	0.983674
6472	GO:0032501	5663.0	1143	719	4944	0.873035
6473	GO:0032502	5606.0	1340	816	4790	0.854442
9330	GO:0044238	4882.0	208	98	4784	0.979926
11080	GO:0048856	5348.0	1162	723	4625	0.864809
2036	GO:0006807	4331.0	45	15	4316	0.996537
14282	GO:0071840	3845.0	322	158	3687	0.958908

```
================================================================================
Top False Negative GO terms: CCO
================================================================================
```

	go_term	true_count	pred_count	tp	fn	fn_rate
176	GO:0005575	18430.0	1819	1462	16968	0.920673
2550	GO:0110165	18117.0	1592	1266	16851	0.930121
212	GO:0005622	14098.0	242	192	13906	0.986381
1554	GO:0043226	12103.0	894	626	11477	0.948277
1557	GO:0043229	11617.0	1102	756	10861	0.934923
278	GO:0005737	10686.0	22	14	10672	0.998690
1555	GO:0043227	11030.0	635	392	10638	0.964461
1559	GO:0043231	10425.0	893	542	9883	0.948010
217	GO:0005634	5737.0	3297	1627	4110	0.716402
641	GO:0016020	5105.0	2335	1166	3939	0.771596
348	GO:0005829	3685.0	66	28	3657	0.992402
1142	GO:0032991	3373.0	434	167	3206	0.950489
2116	GO:0071944	4095.0	2432	1030	3065	0.748474
389	GO:0005886	3345.0	2485	998	2347	0.701644
1045	GO:0031974	3146.0	2795	828	2318	0.736809

```
==========================================================================
====
Top False Negative GO terms: MFO
==========================================================================
====
```

	go_term	true_count	pred_count	tp	fn	fn_rate
275	GO:0003674	15626.0	0	0	15626	1.000000
1409	GO:0005488	11462.0	0	0	11462	1.000000
1421	GO:0005515	9625.0	0	0	9625	1.000000
320	GO:0003824	4961.0	675	407	4554	0.917960
6337	GO:0097159	2523.0	305	109	2414	0.956797
6994	GO:1901363	2481.0	359	124	2357	0.950020
2694	GO:0016740	1959.0	326	107	1852	0.945380
276	GO:0003676	1989.0	817	268	1721	0.865259
2733	GO:0016787	1531.0	289	111	1420	0.927498
6755	GO:0140096	1422.0	269	122	1300	0.914205
4283	GO:0042802	1134.0	0	0	1134	1.000000
4346	GO:0043167	1083.0	225	48	1035	0.955679
277	GO:0003677	1214.0	1156	327	887	0.730643
3128	GO:0019899	782.0	6	1	781	0.998721
6414	GO:0098772	776.0	261	51	725	0.934278

Step 11. Top False Positive GO Terms

Interpretation

False positives are GO terms predicted by the model but not present in the ground truth.

In [12]:
```
for ns in ["BPO", "CCO", "MFO"]:
    print("=" * 80)
    print(f"Top False Positive GO terms: {ns}")
    print("=" * 80)

    top_fp = (
        label_error_tables[ns][label_error_tables[ns]["pred_count"] > 0]
        .sort_values("fp", ascending=False)
        .head(15)
    )

    display(top_fp[["go_term", "true_count", "pred_count", "tp", "fp", "fp_r

    top_fp.to_csv(
        os.path.join(OUTPUT_DIR, f"{ns}_top_false_positive_terms.csv"),
        index=False
    )
```

```
================================================================================
Top False Positive GO terms: BPO
================================================================================
```

	go_term	true_count	pred_count	tp	fp	fp_rate
2090	GO:0006886	284.0	6981	174	6807	0.975075
12516	GO:0060560	254.0	6954	150	6804	0.978430
7180	GO:0034248	250.0	5591	118	5473	0.978895
4453	GO:0016049	244.0	5082	122	4960	0.975994
4212	GO:0015031	416.0	4890	195	4695	0.960123
9634	GO:0045184	458.0	4480	199	4281	0.955580
17922	GO:1902532	220.0	4099	98	4001	0.976092
8381	GO:0040008	495.0	4139	200	3939	0.951679
7295	GO:0034470	270.0	3695	136	3559	0.963194
14687	GO:0072594	189.0	3583	79	3504	0.977951
3760	GO:0010638	198.0	3576	84	3492	0.976510
17104	GO:1901137	419.0	3686	208	3478	0.943570
20561	GO:2000241	270.0	3539	119	3420	0.966375
7716	GO:0035556	550.0	3630	231	3399	0.936364
298	GO:0001558	246.0	3469	90	3379	0.974056

```
================================================================================
Top False Positive GO terms: CCO
================================================================================
```

	go_term	true_count	pred_count	tp	fp	fp_rate
279	GO:0005739	1799.0	4896	994	3902	0.796977
2622	GO:0140513	803.0	3813	407	3406	0.893260
85	GO:0000785	443.0	3564	251	3313	0.929574
236	GO:0005667	242.0	3153	126	3027	0.960038
1039	GO:0031966	355.0	3242	217	3025	0.933066
830	GO:0030659	368.0	3193	169	3024	0.947072
615	GO:0012506	378.0	3145	169	2976	0.946264
280	GO:0005740	385.0	3142	224	2918	0.928708
257	GO:0005694	827.0	3258	413	2845	0.873235
1524	GO:0043005	651.0	2952	295	2657	0.900068
810	GO:0030312	678.0	3027	376	2651	0.875785
404	GO:0005911	423.0	2797	164	2633	0.941366
2393	GO:0098687	237.0	2734	126	2608	0.953914
896	GO:0031012	333.0	2757	196	2561	0.928908
594	GO:0009986	408.0	2735	194	2541	0.929068

```
==============================================================================
====
Top False Positive GO terms: MFO
==============================================================================
====
```

	go_term	true_count	pred_count	tp	fp	fp_rate
5100	GO:0048018	155.0	1828	101	1727	0.944748
2710	GO:0016757	285.0	1806	181	1625	0.899779
3296	GO:0030546	158.0	1673	103	1570	0.938434
3295	GO:0030545	169.0	1638	106	1532	0.935287
3179	GO:0022857	935.0	1969	666	1303	0.661757
1285	GO:0005215	989.0	1929	667	1262	0.654225
3650	GO:0032555	277.0	1326	108	1218	0.918552
2875	GO:0017076	343.0	1258	112	1146	0.910970
56	GO:0000287	123.0	1168	41	1127	0.964897
1211	GO:0005102	555.0	1233	157	1076	0.872668
110	GO:0000981	303.0	1195	171	1024	0.856904
2763	GO:0016829	318.0	1086	116	970	0.893186
3159	GO:0022804	285.0	1162	193	969	0.833907
1420	GO:0005509	142.0	990	51	939	0.948485
2741	GO:0016798	205.0	1053	125	928	0.881292

Step 12. Label Frequency vs False Negative Rate

Why this step?

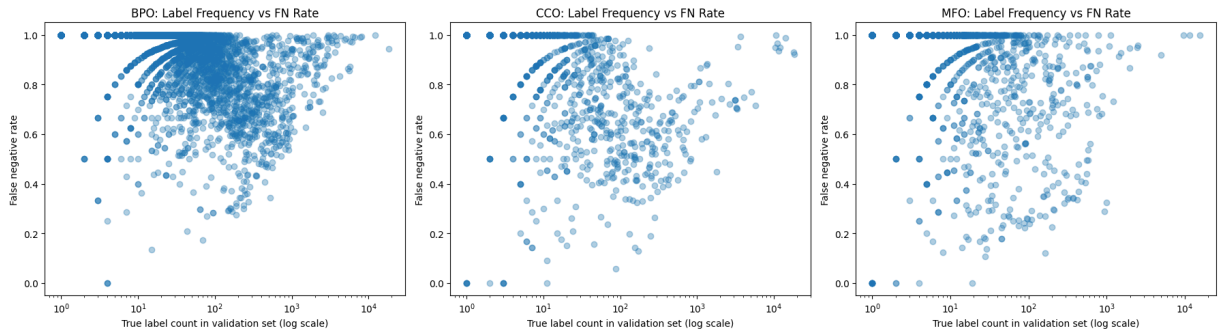
GO terms with fewer positive examples are usually harder to predict.
This plot checks whether rare labels have higher false negative rates.

In [13]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, ns in zip(axes, ["BPO", "CCO", "MFO"]):
    plot_df = label_error_tables[ns].copy()
    plot_df = plot_df[plot_df["true_count"] > 0]

    ax.scatter(plot_df["true_count"], plot_df["fn_rate"], alpha=0.35)
    ax.set_xscale("log")
    ax.set_xlabel("True label count in validation set (log scale)")
    ax.set_ylabel("False negative rate")
    ax.set_title(f"{ns}: Label Frequency vs FN Rate")

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "label_frequency_vs_fn_rate_by_namespac
plt.show()
```



Step 13. Predicted vs True GO Count Distribution by Namespace

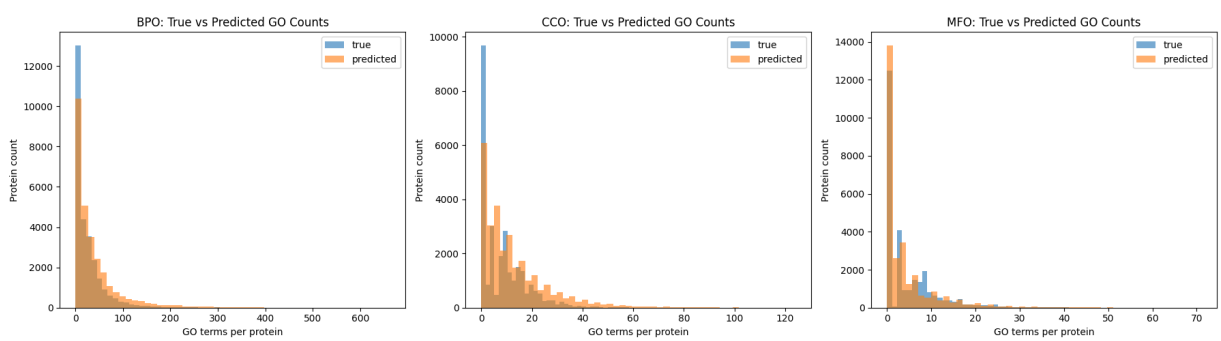
In [14]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, ns in zip(axes, ["BPO", "CCO", "MFO"]):
    y_true = predictions[ns]["y_true"]
    y_prob = predictions[ns]["y_prob"]
    threshold = threshold_results[ns]["selected_threshold"]
    y_pred = (y_prob >= threshold).astype(np.int8)

    true_counts = y_true.sum(axis=1)
    pred_counts = y_pred.sum(axis=1)

    ax.hist(true_counts, bins=50, alpha=0.6, label="true")
    ax.hist(pred_counts, bins=50, alpha=0.6, label="predicted")
    ax.set_title(f"{ns}: True vs Predicted GO Counts")
    ax.set_xlabel("GO terms per protein")
    ax.set_ylabel("Protein count")
    ax.legend()

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "true_vs_predicted_go_count_by_namespac
plt.show()
```



Step 14. Save Combined Error Table

In [15]:
```
combined_label_error_df = pd.concat(
    [label_error_tables[ns] for ns in ["BPO", "CCO", "MFO"]],
    ignore_index=True
)
```

```
combined_label_error_df.to_csv(
    os.path.join(OUTPUT_DIR, "combined_per_label_error_table.csv"),
    index=False
)

summary = {
    "checkpoints": CHECKPOINTS,
    "model_configs_inferred_from_checkpoints": model_configs,
    "threshold_results": threshold_results,
    "error_summary": error_summary_df.to_dict(orient="records"),
}

with open(os.path.join(OUTPUT_DIR, "error_analysis_summary.json"), "w") as f
    json.dump(summary, f, indent=2)

print("Saved all outputs to:", OUTPUT_DIR)
```

```
Saved all outputs to: kmer_error_analysis_outputs
```

Summary

This notebook analyzes the final k-mer GO-split checkpoints:

- `kmer_BPO_final.pt`
- `kmer_CCO_final.pt`
- `kmer_MFO_final.pt`

It does not use `kmer_dropout_0.3.pt`, because that file is an intermediate tuning checkpoint rather than the final namespace-specific model.

The error analysis identifies:

- total TP / FP / FN by namespace
- top false negative GO terms
- top false positive GO terms
- whether rare GO terms tend to have higher false negative rates
- whether each namespace model over-predicts or under-predicts GO labels

ESM GO-split Checkpoint Error Analysis

This notebook performs error analysis for the three namespace-specific ESM MLP checkpoints:

- `esm_BPO_tuned.pt`
- `esm_CCO_tuned.pt`
- `esm_MFO_tuned.pt`

It follows the GO-split setup from `ESM_GO_split_tuning.ipynb`.

This notebook does **not** retrain the models.
It only reloads saved checkpoints, runs validation inference, and analyzes errors.

If you want to analyze the baseline models instead, change the `CHECKPOINTS` dictionary to:

```
CHECKPOINTS = {
    "BPO": "esm_BPO_baseline.pt",
    "CCO": "esm_CCO_baseline.pt",
    "MFO": "esm_MFO_baseline.pt",
}
```

Step 0. Imports and Paths

In [2]:
```
import os
import json
import pickle
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.sparse import load_npz

import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from sklearn.metrics import f1_score
from sklearn.preprocessing import StandardScaler

DATA_DIR = "data_processed_esm"
KMER_DATA_DIR = "data_processed_kmer"

# Default: analyze tuned GO-split ESM models
CHECKPOINTS = {
    "BPO": "pt/esm_BPO_tuned.pt",
    "CCO": "pt/esm_CCO_tuned.pt",
    "MFO": "pt/esm_MFO_tuned.pt",
```

```
}

# If you want baseline instead, use:
# CHECKPOINTS = {
#     "BPO": "esm_BPO_baseline.pt",
#     "CCO": "esm_CCO_baseline.pt",
#     "MFO": "esm_MFO_baseline.pt",
# }

OUTPUT_DIR = "esm_error_analysis_outputs"
os.makedirs(OUTPUT_DIR, exist_ok=True)

print("DATA_DIR:", DATA_DIR)
print("KMER_DATA_DIR:", KMER_DATA_DIR)
print("CHECKPOINTS:", CHECKPOINTS)
print("OUTPUT_DIR:", OUTPUT_DIR)
```

```
DATA_DIR: data_processed_esm
KMER_DATA_DIR: data_processed_kmer
CHECKPOINTS: {'BPO': 'pt/esm_BPO_tuned.pt', 'CCO': 'pt/esm_CCO_tuned.pt', 'M
FO': 'pt/esm_MFO_tuned.pt'}
OUTPUT_DIR: esm_error_analysis_outputs
```

Step 1. Load Processed ESM Embeddings and Labels

Why this step?

The `.pt` checkpoint only stores model weights.
To evaluate errors, we need the ESM validation embeddings and the corresponding GO labels.

We reuse the same `train_idx.npy` and `val_idx.npy` from the k-mer pipeline so that comparisons remain aligned across k-mer and ESM experiments.

In [3]:
```
X_esm = np.load(os.path.join(DATA_DIR, "X_esm.npy"))
print("X_esm:", X_esm.shape, X_esm.dtype)

y_sparse_path = os.path.join(DATA_DIR, "Y_sparse.npz")
y_dense_path = os.path.join(DATA_DIR, "Y.npy")

if os.path.exists(y_sparse_path):
    Y = load_npz(y_sparse_path)
    print("Loaded sparse Y:", Y.shape, type(Y))
elif os.path.exists(y_dense_path):
    Y = np.load(y_dense_path)
    print("Loaded dense Y:", Y.shape, type(Y))
else:
    raise FileNotFoundError("Cannot find Y_sparse.npz or Y.npy in data_proce

train_idx = np.load(os.path.join(KMER_DATA_DIR, "train_idx.npy"))
val_idx = np.load(os.path.join(KMER_DATA_DIR, "val_idx.npy"))

X_train = X_esm[train_idx]
```

```
X_val = X_esm[val_idx]

if hasattr(Y, "tocsr"):
    Y_train = Y[train_idx]
    Y_val = Y[val_idx]
else:
    Y_train = Y[train_idx]
    Y_val = Y[val_idx]

print("X_train:", X_train.shape)
print("X_val:  ", X_val.shape)
print("Y_train:", Y_train.shape)
print("Y_val:  ", Y_val.shape)

assert X_val.shape[0] == Y_val.shape[0]
```

```
X_esm: (140569, 640) float32
Loaded sparse Y: (140569, 31454) <class 'scipy.sparse._csr.csr_matrix'>
X_train: (112455, 640)
X_val:   (28114, 640)
Y_train: (112455, 31454)
Y_val:   (28114, 31454)
```

Step 2. Standardize ESM Features

Why this step?

The ESM GO-split training notebook standardized the embeddings using training-set statistics.
We must repeat the same transformation before loading the trained checkpoint.

If you already saved `esm_scaler.pkl`, this cell will load it.
Otherwise, it fits a scaler on `X_train` and applies it to `X_val`.

In [4]:
```
scaler_path = "esm_scaler.pkl"

if os.path.exists(scaler_path):
    with open(scaler_path, "rb") as f:
        scaler = pickle.load(f)
    X_train = scaler.transform(X_train).astype(np.float32)
    X_val = scaler.transform(X_val).astype(np.float32)
    print("Loaded existing scaler from:", scaler_path)
else:
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train).astype(np.float32)
    X_val = scaler.transform(X_val).astype(np.float32)
    with open(scaler_path, "wb") as f:
        pickle.dump(scaler, f)
    print("Fitted and saved scaler to:", scaler_path)

print("Standardized X_train:", X_train.shape, X_train.dtype)
print("Standardized X_val:  ", X_val.shape, X_val.dtype)
```

```
Loaded existing scaler from: esm_scaler.pkl
Standardized X_train: (112455, 640) float32
Standardized X_val:   (28114, 640) float32
```

Step 3. Load GO Term Order

Why this step?

To split labels into BPO / CCO / MFO, we need to know which GO term each column of `Y` represents.

For the ESM pipeline, this is usually stored in `mlb.pkl`.

In [5]:
```
mlb_path = os.path.join(DATA_DIR, "mlb.pkl")

if os.path.exists(mlb_path):
    with open(mlb_path, "rb") as f:
        mlb = pickle.load(f)
    go_terms = list(mlb.classes_)
    print("Loaded GO terms from mlb.pkl")
else:
    meta_path = os.path.join(DATA_DIR, "meta.pkl")
    if os.path.exists(meta_path):
        with open(meta_path, "rb") as f:
            meta = pickle.load(f)

        if isinstance(meta, dict) and "go_terms" in meta:
            go_terms = meta["go_terms"].tolist() if hasattr(meta["go_terms"]
            print("Loaded GO terms from meta['go_terms']")
        elif hasattr(meta, "classes_"):
            go_terms = list(meta.classes_)
            print("Loaded GO terms from meta.classes_")
        else:
            raise KeyError("Could not find GO term order in meta.pkl")
    else:
        raise FileNotFoundError("Could not find mlb.pkl or meta.pkl in data_

print("Number of GO terms:", len(go_terms))
print("Y columns:", Y_val.shape[1])
assert len(go_terms) == Y_val.shape[1]
```

```
Loaded GO terms from mlb.pkl
Number of GO terms: 31454
Y columns: 31454
```

Step 4. Load GO Namespace Mapping

Why this step?

The GO-split models were trained separately for:

- BPO

- CCO
- MFO

We reconstruct the same column splits using `train_terms.tsv`.

In [6]:
```
TERMS_PATH = os.path.join("..", "cafa-5-protein-function-prediction", "Trair

if not os.path.exists(TERMS_PATH):
    raise FileNotFoundError(
        f"Cannot find train_terms.tsv at {TERMS_PATH}. "
        "Please update TERMS_PATH."
    )

terms_df = pd.read_csv(TERMS_PATH, sep="\t")

go_to_aspect = (
    terms_df[["term", "aspect"]]
    .drop_duplicates()
    .set_index("term")["aspect"]
    .to_dict()
)

bpo_idx = [i for i, go in enumerate(go_terms) if go_to_aspect.get(go) == "BF
cco_idx = [i for i, go in enumerate(go_terms) if go_to_aspect.get(go) == "CC
mfo_idx = [i for i, go in enumerate(go_terms) if go_to_aspect.get(go) == "MF

namespace_indices = {
    "BPO": bpo_idx,
    "CCO": cco_idx,
    "MFO": mfo_idx,
}

namespace_go_terms = {
    "BPO": [go_terms[i] for i in bpo_idx],
    "CCO": [go_terms[i] for i in cco_idx],
    "MFO": [go_terms[i] for i in mfo_idx],
}

print("Total GO terms:", len(go_terms))
print("BPO labels:", len(bpo_idx))
print("CCO labels:", len(cco_idx))
print("MFO labels:", len(mfo_idx))
print("Check total:", len(bpo_idx) + len(cco_idx) + len(mfo_idx))
```

```
Total GO terms: 31454
BPO labels: 21279
CCO labels: 2957
MFO labels: 7218
Check total: 31454
```

Step 5. Split Validation Labels by Namespace

In [7]:
```
Y_val_namespace = {
    "BPO": Y_val[:, bpo_idx],
```

```
    "CCO": Y_val[:, cco_idx],
    "MFO": Y_val[:, mfo_idx],
}

Y_train_namespace = {
    "BPO": Y_train[:, bpo_idx],
    "CCO": Y_train[:, cco_idx],
    "MFO": Y_train[:, mfo_idx],
}

for ns in ["BPO", "CCO", "MFO"]:
    print(ns, "Y_train:", Y_train_namespace[ns].shape, "Y_val:", Y_val_names
```

```
BPO Y_train: (112455, 21279) Y_val: (28114, 21279)
CCO Y_train: (112455, 2957) Y_val: (28114, 2957)
MFO Y_train: (112455, 7218) Y_val: (28114, 7218)
```

Step 6. Define ESM Dataset and MLP Architecture

Why this step?

Each `.pt` checkpoint stores model weights only, so we rebuild the same ESM MLP architecture before loading each checkpoint.

In [8]:
```
class ProteinESMDataset(Dataset):
    def __init__(self, X, Y):
        self.X = X.astype(np.float32)
        self.Y = Y
        self.is_sparse = hasattr(Y, "tocsr")
        if self.is_sparse:
            self.Y = Y.tocsr()
            self.n_labels = self.Y.shape[1]

    def __len__(self):
        return self.X.shape[0]

    def __getitem__(self, idx):
        x = torch.from_numpy(self.X[idx])

        if self.is_sparse:
            start = self.Y.indptr[idx]
            end = self.Y.indptr[idx + 1]
            pos_idx = self.Y.indices[start:end]

            y = torch.zeros(self.n_labels, dtype=torch.float32)
            y[pos_idx] = 1.0
        else:
            y = torch.tensor(self.Y[idx], dtype=torch.float32)

        return x, y


class ESMMLP(nn.Module):
    def __init__(self, input_dim, output_dim, hidden1=512, hidden2=256, drop
```

```
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden1),
            nn.BatchNorm1d(hidden1),
            nn.ReLU(),
            nn.Dropout(dropout),
            nn.Linear(hidden1, hidden2),
            nn.BatchNorm1d(hidden2),
            nn.ReLU(),
            nn.Dropout(dropout),
            nn.Linear(hidden2, output_dim)
        )

    def forward(self, x):
        return self.net(x)


def build_val_loader(X_val, Y_val_ns, batch_size=64):
    dataset = ProteinESMDataset(X_val, Y_val_ns)
    loader = DataLoader(
        dataset,
        batch_size=batch_size,
        shuffle=False,
        num_workers=4,
        pin_memory=torch.cuda.is_available()
    )
    return dataset, loader


device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
input_dim = X_val.shape[1]

print("Device:", device)
print("input_dim:", input_dim)
```

```
Device: cpu
input_dim: 640
```

```
/global/home/users/yushanfu/esm_env/lib/python3.11/site-packages/torch/cuda/
__init__.py:180: UserWarning: CUDA initialization: The NVIDIA driver on your
system is too old (found version 12030). Please update your GPU driver by do
wnloading and installing a new version from the URL: http://www.nvidia.com/D
ownload/index.aspx Alternatively, go to: https://pytorch.org to install a Py
Torch version that has been compiled with your version of the CUDA driver.
(Triggered internally at /pytorch/c10/cuda/CUDAFunctions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
```

Step 7. Load Checkpoint and Infer Architecture

Why this step?

If the saved model uses hidden sizes different from the default, we infer the hidden dimensions from the checkpoint weight shapes.

In [9]:
```
def load_checkpoint_state_dict(path):
    if not os.path.exists(path):
        raise FileNotFoundError(
            f"Checkpoint not found: {path}. "
            "Please make sure the .pt file is in the current directory."
        )

    checkpoint = torch.load(path, map_location="cpu")

    if isinstance(checkpoint, dict) and "model_state_dict" in checkpoint:
        return checkpoint["model_state_dict"]
    elif isinstance(checkpoint, dict):
        return checkpoint
    else:
        raise ValueError("Unsupported checkpoint format.")


def infer_mlp_dims_from_state_dict(state_dict):
    if "net.0.weight" in state_dict and "net.4.weight" in state_dict:
        hidden1 = state_dict["net.0.weight"].shape[0]
        hidden2 = state_dict["net.4.weight"].shape[0]
    else:
        linear_weight_keys = [
            k for k in state_dict.keys()
            if k.endswith("weight") and len(state_dict[k].shape) == 2
        ]
        print("Detected linear-like weight keys:")
        for k in linear_weight_keys:
            print(k, tuple(state_dict[k].shape))

        hidden1 = state_dict[linear_weight_keys[0]].shape[0]
        hidden2 = state_dict[linear_weight_keys[1]].shape[0]

    return int(hidden1), int(hidden2)


def build_model_from_checkpoint(checkpoint_path, input_dim, output_dim, devi
    state_dict = load_checkpoint_state_dict(checkpoint_path)
    hidden1, hidden2 = infer_mlp_dims_from_state_dict(state_dict)

    model = ESMMLP(
        input_dim=input_dim,
        output_dim=output_dim,
        hidden1=hidden1,
        hidden2=hidden2,
        dropout=dropout
    ).to(device)

    model.load_state_dict(state_dict)
    model.eval()

    return model, {"hidden1": hidden1, "hidden2": hidden2, "dropout": dropou
```

Step 8. Run Validation Inference for Each Namespace

Why this step?

We only need a forward pass through the validation set.

No training is performed here.

In [10]:
```
@torch.no_grad()
def get_predictions(model, loader, device):
    model.eval()
    all_probs = []
    all_targets = []

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        logits = model(X_batch)
        probs = torch.sigmoid(logits).cpu().numpy()

        all_probs.append(probs)
        all_targets.append(Y_batch.numpy())

    y_prob = np.vstack(all_probs)
    y_true = np.vstack(all_targets)

    return y_prob, y_true


predictions = {}
model_configs = {}

for ns in ["BPO", "CCO", "MFO"]:
    print("=" * 80)
    print(f"Loading and evaluating {ns}")
    print("=" * 80)

    _, val_loader = build_val_loader(X_val, Y_val_namespace[ns], batch_size=

    model, inferred_config = build_model_from_checkpoint(
        checkpoint_path=CHECKPOINTS[ns],
        input_dim=input_dim,
        output_dim=Y_val_namespace[ns].shape[1],
        device=device,
        dropout=0.3
    )

    y_prob, y_true = get_predictions(model, val_loader, device)

    predictions[ns] = {
        "y_prob": y_prob,
        "y_true": y_true,
    }

    model_configs[ns] = inferred_config

    np.save(os.path.join(OUTPUT_DIR, f"{ns}_y_prob.npy"), y_prob)
    np.save(os.path.join(OUTPUT_DIR, f"{ns}_y_true.npy"), y_true)
```

```
    print(ns, "y_prob:", y_prob.shape, "y_true:", y_true.shape)
    print(ns, "inferred config:", inferred_config)
```

```
==============================================================================
====
Loading and evaluating BPO
==============================================================================
====
BPO y_prob: (28114, 21279) y_true: (28114, 21279)
BPO inferred config: {'hidden1': 1024, 'hidden2': 512, 'dropout': 0.3}
==============================================================================
====
Loading and evaluating CCO
==============================================================================
====
CCO y_prob: (28114, 2957) y_true: (28114, 2957)
CCO inferred config: {'hidden1': 1024, 'hidden2': 512, 'dropout': 0.3}
==============================================================================
====
Loading and evaluating MFO
==============================================================================
====
MFO y_prob: (28114, 7218) y_true: (28114, 7218)
MFO inferred config: {'hidden1': 1024, 'hidden2': 512, 'dropout': 0.3}
```

Step 9. Threshold Selection

Why this step?

Error analysis requires binary predictions.

If you already know the best thresholds from your previous ESM GO-split notebook, fill them in `BEST_THRESHOLDS`.

If you leave a value as `None`, the notebook will re-run threshold sweeping for that namespace.

In [11]:
```
# Fill in your known thresholds here if you already have them.
# Example:
# BEST_THRESHOLDS = {"BPO": 0.75, "CCO": 0.65, "MFO": 0.85}

BEST_THRESHOLDS = {
    "BPO": None,
    "CCO": None,
    "MFO": None,
}


def threshold_sweep(y_true, y_prob, thresholds=None):
    if thresholds is None:
        thresholds = np.arange(0.05, 0.95, 0.05)
```

```
    records = []

    for t in thresholds:
        y_pred = (y_prob >= t).astype(np.int8)
        micro = f1_score(y_true, y_pred, average="micro", zero_division=0)
        macro = f1_score(y_true, y_pred, average="macro", zero_division=0)

        records.append({
            "threshold": float(t),
            "micro_f1": float(micro),
            "macro_f1": float(macro),
        })

    best_micro = max(records, key=lambda x: x["micro_f1"])
    best_macro = max(records, key=lambda x: x["macro_f1"])

    return records, best_micro, best_macro


threshold_results = {}

for ns in ["BPO", "CCO", "MFO"]:
    y_prob = predictions[ns]["y_prob"]
    y_true = predictions[ns]["y_true"]

    if BEST_THRESHOLDS[ns] is not None:
        selected_threshold = float(BEST_THRESHOLDS[ns])
        y_pred = (y_prob >= selected_threshold).astype(np.int8)
        micro = f1_score(y_true, y_pred, average="micro", zero_division=0)
        macro = f1_score(y_true, y_pred, average="macro", zero_division=0)

        threshold_results[ns] = {
            "records": None,
            "best_micro": {
                "threshold": selected_threshold,
                "micro_f1": float(micro),
                "macro_f1": float(macro),
            },
            "best_macro": None,
            "selected_threshold": selected_threshold,
        }

        print("=" * 80)
        print(ns)
        print("Using provided threshold:", selected_threshold)
        print("micro-F1:", micro)
        print("macro-F1:", macro)

    else:
        records, best_micro, best_macro = threshold_sweep(y_true, y_prob)

        threshold_results[ns] = {
            "records": records,
            "best_micro": best_micro,
            "best_macro": best_macro,
            "selected_threshold": best_micro["threshold"],
```

```
        }

        print("=" * 80)
        print(ns)
        print("Best micro:", best_micro)
        print("Best macro:", best_macro)

        pd.DataFrame(records).to_csv(
            os.path.join(OUTPUT_DIR, f"{ns}_threshold_sweep.csv"),
            index=False
        )
```

```
================================================================================
====
BPO
Best micro: {'threshold': 0.8, 'micro_f1': 0.1843773452426078, 'macro_f1':
0.04060551508860181}
Best macro: {'threshold': 0.6500000000000001, 'micro_f1': 0.153561399642107
1, 'macro_f1': 0.044144412197699004}
================================================================================
====
CCO
Best micro: {'threshold': 0.6500000000000001, 'micro_f1': 0.262879565728471
6, 'macro_f1': 0.07836417271972405}
Best macro: {'threshold': 0.7500000000000001, 'micro_f1': 0.250688261358743
2, 'macro_f1': 0.08315604591051522}
================================================================================
====
MFO
Best micro: {'threshold': 0.9000000000000001, 'micro_f1': 0.285033275286251
2, 'macro_f1': 0.09184794198637769}
Best macro: {'threshold': 0.8, 'micro_f1': 0.25901231625129023, 'macro_f1':
0.11046465335130193}
```

Step 10. Compute FP / FN / TP Error Summary

In [12]:
```
def compute_error_analysis_for_namespace(ns, y_true, y_prob, threshold, go_t
    y_pred = (y_prob >= threshold).astype(np.int8)

    fp = (y_pred == 1) & (y_true == 0)
    fn = (y_pred == 0) & (y_true == 1)
    tp = (y_pred == 1) & (y_true == 1)

    fp_per_label = fp.sum(axis=0)
    fn_per_label = fn.sum(axis=0)
    tp_per_label = tp.sum(axis=0)

    true_count_per_label = y_true.sum(axis=0)
    pred_count_per_label = y_pred.sum(axis=0)

    summary = {
        "namespace": ns,
        "threshold": float(threshold),
        "total_tp": int(tp.sum()),
        "total_fp": int(fp.sum()),
```

```
        "total_fn": int(fn.sum()),
        "total_true_labels": int(y_true.sum()),
        "total_predicted_labels": int(y_pred.sum()),
        "avg_true_labels_per_protein": float(y_true.sum(axis=1).mean()),
        "avg_predicted_labels_per_protein": float(y_pred.sum(axis=1).mean())
    }

    label_error_df = pd.DataFrame({
        "namespace": ns,
        "go_term": go_terms_ns,
        "true_count": true_count_per_label,
        "pred_count": pred_count_per_label,
        "tp": tp_per_label,
        "fp": fp_per_label,
        "fn": fn_per_label,
    })

    label_error_df["fn_rate"] = label_error_df["fn"] / (label_error_df["true
    label_error_df["fp_rate"] = label_error_df["fp"] / (label_error_df["prec

    return summary, label_error_df


error_summaries = []
label_error_tables = {}

for ns in ["BPO", "CCO", "MFO"]:
    y_true = predictions[ns]["y_true"]
    y_prob = predictions[ns]["y_prob"]
    threshold = threshold_results[ns]["selected_threshold"]

    summary, label_error_df = compute_error_analysis_for_namespace(
        ns=ns,
        y_true=y_true,
        y_prob=y_prob,
        threshold=threshold,
        go_terms_ns=namespace_go_terms[ns],
    )

    error_summaries.append(summary)
    label_error_tables[ns] = label_error_df

    label_error_df.to_csv(
        os.path.join(OUTPUT_DIR, f"{ns}_per_label_error_table.csv"),
        index=False
    )

error_summary_df = pd.DataFrame(error_summaries)
display(error_summary_df)

error_summary_df.to_csv(
    os.path.join(OUTPUT_DIR, "namespace_error_summary.csv"),
    index=False
)
```

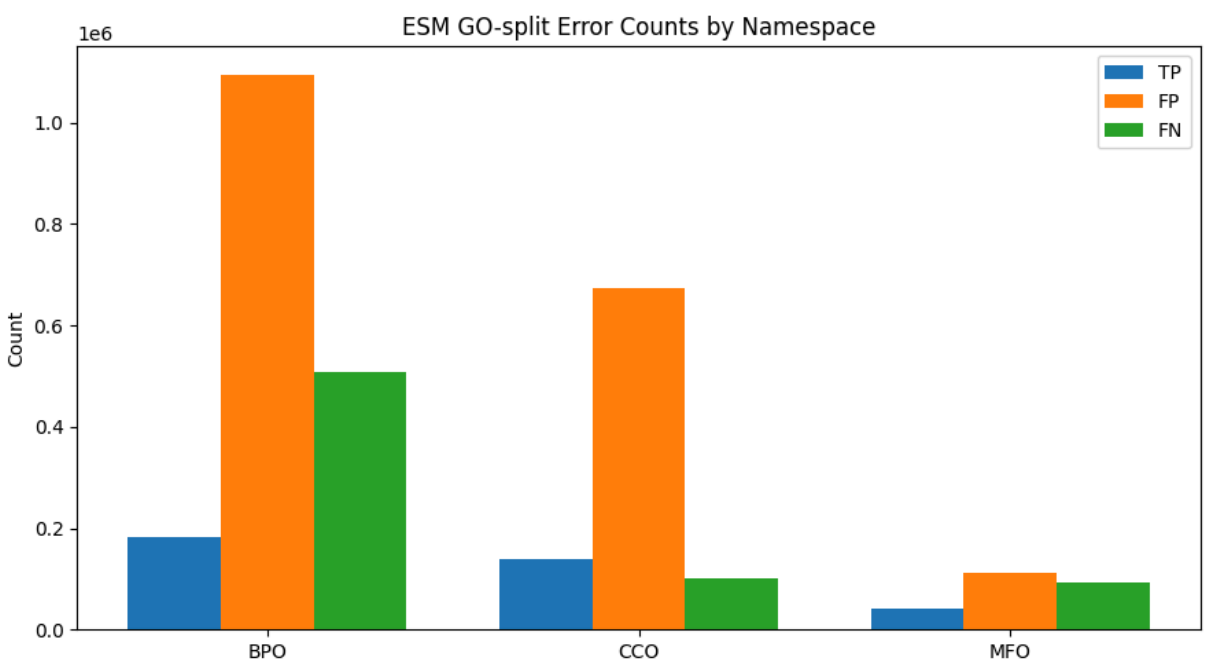
	namespace	threshold	total_tp	total_fp	total_fn	total_true_labels	total_predicted
0	BPO	0.80	181189	1094857	508180	689369	1
1	CCO	0.65	137871	672665	100522	238393	
2	MFO	0.90	40838	112345	92528	133366	

Step 11. Plot Error Counts by Namespace

In [13]:
```
x = np.arange(len(error_summary_df))
width = 0.25

plt.figure(figsize=(9, 5))
plt.bar(x - width, error_summary_df["total_tp"], width=width, label="TP")
plt.bar(x, error_summary_df["total_fp"], width=width, label="FP")
plt.bar(x + width, error_summary_df["total_fn"], width=width, label="FN")

plt.xticks(x, error_summary_df["namespace"])
plt.ylabel("Count")
plt.title("ESM GO-split Error Counts by Namespace")
plt.legend()
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "namespace_error_counts.png"), dpi=200,
plt.show()
```



Step 12. Top False Negative GO Terms

Interpretation

False negatives are GO terms that are truly present but missed by the model.

In [14]:
```
for ns in ["BPO", "CCO", "MFO"]:
    print("=" * 80)
    print(f"Top False Negative GO terms: {ns}")
    print("=" * 80)

    top_fn = (
        label_error_tables[ns][label_error_tables[ns]["true_count"] > 0]
        .sort_values("fn", ascending=False)
        .head(15)
    )

    display(top_fn[["go_term", "true_count", "pred_count", "tp", "fn", "fn_r

    top_fn.to_csv(
        os.path.join(OUTPUT_DIR, f"{ns}_top_false_negative_terms.csv"),
        index=False
    )
```

```
================================================================================
Top False Negative GO terms: BPO
================================================================================
```

	go_term	true_count	pred_count	tp	fn	fn_rate
2653	GO:0008150	18271.0	2701	2353	15918	0.871217
3304	GO:0009987	12184.0	15	9	12175	0.999261
13362	GO:0065007	8153.0	744	541	7612	0.933644
11212	GO:0050789	7699.0	886	641	7058	0.916742
11216	GO:0050794	6617.0	1129	718	5899	0.891492
11286	GO:0050896	6144.0	523	319	5825	0.948079
2654	GO:0008152	6106.0	1773	1060	5046	0.826400
14224	GO:0071704	5657.0	1984	1136	4521	0.799187
6472	GO:0032501	5663.0	2262	1380	4283	0.756313
9330	GO:0044238	4882.0	1238	672	4210	0.862351
6473	GO:0032502	5606.0	2336	1459	4147	0.739743
9329	GO:0044237	5084.0	1830	976	4108	0.808025
11080	GO:0048856	5348.0	2397	1464	3884	0.726253
2036	GO:0006807	4331.0	1274	647	3684	0.850612
14282	GO:0071840	3845.0	974	477	3368	0.875943

```
================================================================================
Top False Negative GO terms: CCO
================================================================================
```

	go_term	true_count	pred_count	tp	fn	fn_rate
212	GO:0005622	14098.0	8136	6175	7923	0.561995
1554	GO:0043226	12103.0	7878	5482	6621	0.547054
176	GO:0005575	18430.0	15564	11840	6590	0.357569
278	GO:0005737	10686.0	5937	4113	6573	0.615104
2550	GO:0110165	18117.0	15287	11560	6557	0.361925
1555	GO:0043227	11030.0	7251	4796	6234	0.565186
1557	GO:0043229	11617.0	7888	5421	6196	0.533356
1559	GO:0043231	10425.0	7351	4814	5611	0.538225
217	GO:0005634	5737.0	7150	3587	2150	0.374760
348	GO:0005829	3685.0	4766	1576	2109	0.572320
641	GO:0016020	5105.0	6958	3139	1966	0.385113
1142	GO:0032991	3373.0	4873	1564	1809	0.536318
2116	GO:0071944	4095.0	5898	2440	1655	0.404151
1045	GO:0031974	3146.0	6472	1914	1232	0.391608
1561	GO:0043233	3145.0	6479	1916	1229	0.390779

```
==============================================================================
====
Top False Negative GO terms: MFO
==============================================================================
====
```

	go_term	true_count	pred_count	tp	fn	fn_rate
275	GO:0003674	15626.0	1	1	15625	0.999936
1409	GO:0005488	11462.0	0	0	11462	1.000000
1421	GO:0005515	9625.0	0	0	9625	1.000000
320	GO:0003824	4961.0	997	682	4279	0.862528
6337	GO:0097159	2523.0	945	437	2086	0.826793
6994	GO:1901363	2481.0	960	445	2036	0.820637
276	GO:0003676	1989.0	1442	620	1369	0.688286
2694	GO:0016740	1959.0	1273	719	1240	0.632976
4283	GO:0042802	1134.0	1	1	1133	0.999118
4346	GO:0043167	1083.0	168	57	1026	0.947368
2733	GO:0016787	1531.0	1113	593	938	0.612671
6755	GO:0140096	1422.0	1114	550	872	0.613221
3128	GO:0019899	782.0	66	13	769	0.983376
6414	GO:0098772	776.0	624	214	562	0.724227
277	GO:0003677	1214.0	1655	663	551	0.453871

Step 13. Top False Positive GO Terms

Interpretation

False positives are GO terms predicted by the model but not present in the ground truth.

In [15]:
```
for ns in ["BPO", "CCO", "MFO"]:
    print("=" * 80)
    print(f"Top False Positive GO terms: {ns}")
    print("=" * 80)

    top_fp = (
        label_error_tables[ns][label_error_tables[ns]["pred_count"] > 0]
        .sort_values("fp", ascending=False)
        .head(15)
    )

    display(top_fp[["go_term", "true_count", "pred_count", "tp", "fp", "fp_r

    top_fp.to_csv(
        os.path.join(OUTPUT_DIR, f"{ns}_top_false_positive_terms.csv"),
        index=False
    )
```

```
================================================================================
Top False Positive GO terms: BPO
================================================================================
```

	go_term	true_count	pred_count	tp	fp	fp_rate
9354	GO:0044283	537.0	3007	411	2596	0.863319
10262	GO:0046394	339.0	2763	253	2510	0.908433
4457	GO:0016053	342.0	2749	254	2495	0.907603
17337	GO:1901566	1030.0	2854	583	2271	0.795725
4589	GO:0016477	504.0	2462	272	2190	0.889521
10992	GO:0048729	569.0	2437	293	2144	0.879770
17359	GO:1901615	365.0	2322	208	2114	0.910422
1511	GO:0006082	891.0	2709	625	2084	0.769288
5149	GO:0019752	845.0	2681	597	2084	0.777322
9079	GO:0043436	887.0	2685	616	2069	0.770577
540	GO:0002009	475.0	2309	240	2069	0.896059
9352	GO:0044281	1500.0	3110	1042	2068	0.664952
8872	GO:0043009	443.0	2233	203	2030	0.909091
2530	GO:0007507	434.0	2179	184	1995	0.915558
11092	GO:0048870	610.0	2247	270	1977	0.879840

```
================================================================================
Top False Positive GO terms: CCO
================================================================================
```

	go_term	true_count	pred_count	tp	fp	fp_rate
683	GO:0016604	301.0	5526	226	5300	0.959102
1936	GO:0070062	441.0	5229	311	4918	0.940524
274	GO:0005730	659.0	5317	481	4836	0.909535
1799	GO:0048471	238.0	4886	126	4760	0.974212
1050	GO:0031981	2673.0	6395	1764	4631	0.724159
1667	GO:0044297	329.0	4798	199	4599	0.958524
1558	GO:0043230	471.0	4888	317	4571	0.935147
226	GO:0005654	2054.0	5955	1384	4571	0.767590
1928	GO:0065010	470.0	4887	317	4570	0.935134
1561	GO:0043233	3145.0	6479	1916	4563	0.704275
1929	GO:0070013	3142.0	6477	1915	4562	0.704338
2695	GO:1903561	454.0	4866	305	4561	0.937320
1051	GO:0031982	1434.0	5397	836	4561	0.845099
1045	GO:0031974	3146.0	6472	1914	4558	0.704265
1526	GO:0043025	289.0	4679	174	4505	0.962813

```
==============================================================================
====
Top False Positive GO terms: MFO
==============================================================================
====
```

	go_term	true_count	pred_count	tp	fp	fp_rate
131	GO:0001067	570.0	1657	406	1251	0.754979
106	GO:0000976	568.0	1632	402	1230	0.753676
291	GO:0003700	581.0	1653	460	1193	0.721718
7190	GO:1990837	657.0	1637	457	1180	0.720831
285	GO:0003690	741.0	1630	461	1169	0.717178
4048	GO:0035639	235.0	1258	132	1126	0.895072
111	GO:0000987	210.0	1265	147	1118	0.883794
107	GO:0000977	246.0	1290	189	1101	0.853488
4398	GO:0043565	871.0	1652	584	1068	0.646489
6764	GO:0140110	760.0	1441	441	1000	0.693963
277	GO:0003677	1214.0	1655	663	992	0.599396
108	GO:0000978	170.0	1091	122	969	0.888176
3650	GO:0032555	277.0	1095	139	956	0.873059
3648	GO:0032553	298.0	1091	137	954	0.874427
110	GO:0000981	303.0	1200	247	953	0.794167

Step 14. Label Frequency vs False Negative Rate

Why this step?

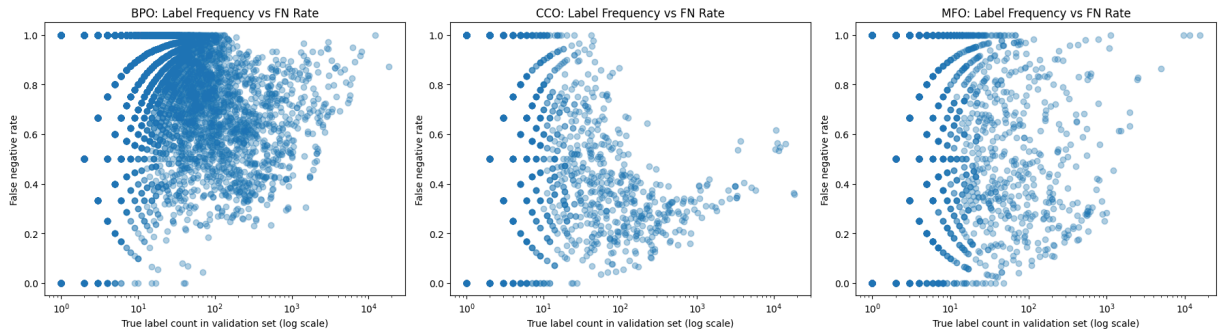
GO terms with fewer positive examples are usually harder to predict.
This plot checks whether rare labels have higher false negative rates.

In [16]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, ns in zip(axes, ["BPO", "CCO", "MFO"]):
    plot_df = label_error_tables[ns].copy()
    plot_df = plot_df[plot_df["true_count"] > 0]

    ax.scatter(plot_df["true_count"], plot_df["fn_rate"], alpha=0.35)
    ax.set_xscale("log")
    ax.set_xlabel("True label count in validation set (log scale)")
    ax.set_ylabel("False negative rate")
    ax.set_title(f"{ns}: Label Frequency vs FN Rate")

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "label_frequency_vs_fn_rate_by_namespac
plt.show()
```



Step 15. Predicted vs True GO Count Distribution by Namespace

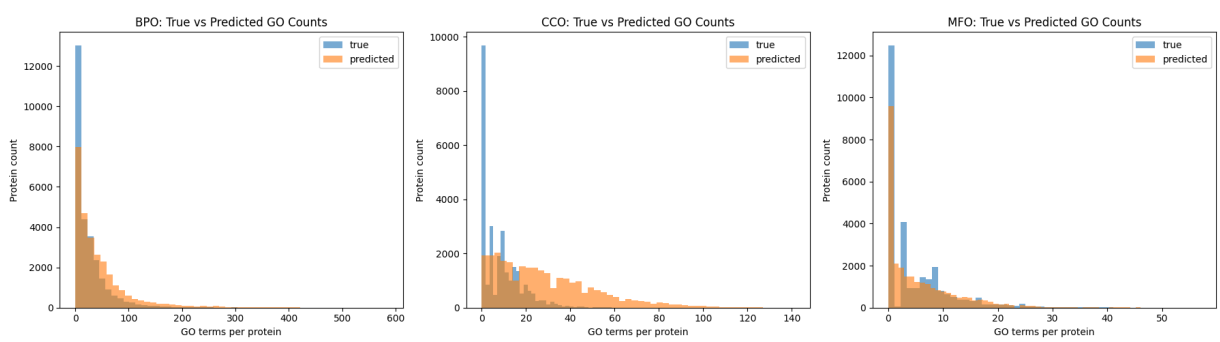
In [17]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, ns in zip(axes, ["BPO", "CCO", "MFO"]):
    y_true = predictions[ns]["y_true"]
    y_prob = predictions[ns]["y_prob"]
    threshold = threshold_results[ns]["selected_threshold"]
    y_pred = (y_prob >= threshold).astype(np.int8)

    true_counts = y_true.sum(axis=1)
    pred_counts = y_pred.sum(axis=1)

    ax.hist(true_counts, bins=50, alpha=0.6, label="true")
    ax.hist(pred_counts, bins=50, alpha=0.6, label="predicted")
    ax.set_title(f"{ns}: True vs Predicted GO Counts")
    ax.set_xlabel("GO terms per protein")
    ax.set_ylabel("Protein count")
    ax.legend()

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, "true_vs_predicted_go_count_by_namespac
plt.show()
```



Step 16. Save Combined Error Table

In [18]:
```
combined_label_error_df = pd.concat(
    [label_error_tables[ns] for ns in ["BPO", "CCO", "MFO"]],
    ignore_index=True
)
```

```
combined_label_error_df.to_csv(
    os.path.join(OUTPUT_DIR, "combined_per_label_error_table.csv"),
    index=False
)

summary = {
    "checkpoints": CHECKPOINTS,
    "model_configs_inferred_from_checkpoints": model_configs,
    "threshold_results": threshold_results,
    "error_summary": error_summary_df.to_dict(orient="records"),
}

with open(os.path.join(OUTPUT_DIR, "error_analysis_summary.json"), "w") as f
    json.dump(summary, f, indent=2)

print("Saved all outputs to:", OUTPUT_DIR)
```

```
Saved all outputs to: esm_error_analysis_outputs
```

Summary

This notebook analyzes the final ESM GO-split checkpoints:

- `esm_BPO_tuned.pt`
- `esm_CCO_tuned.pt`
- `esm_MFO_tuned.pt`

It does not retrain the models.
It loads saved checkpoints, reruns validation inference, selects or reuses thresholds, and computes error analysis.

The error analysis identifies:

- total TP / FP / FN by namespace
- top false negative GO terms
- top false positive GO terms
- whether rare GO terms tend to have higher false negative rates
- whether each namespace model over-predicts or under-predicts GO labels

In [ ]:

```
```